03 21826 Autumn 2025
LH The Life and Death of Stars

Lecture 6

(JCD 4.4 - 4.6)

Annelies Mortier - office PW G35 - a.mortier@bham.ac.uk

UNIVERSITY OF BIRMINGHAM

Previous lecture

- Mass equation:

$$\frac{dm(r)}{dr} = \frac{dm}{dr} = 4\pi r^2 \rho(r)$$

- Equation of hydrostatic equilibrium:

$$\frac{dP(r)}{dr} = -\frac{GM(r)\rho(r)}{r^2}$$

- Estimates of central pressure and temperature from basic physics principles not far of the modeled values
- Radiation pressure more important for massive stars

Slido

Join SLIDO - #1114155

Outline

The virial theorem

Simple models

Polytropes

Summary

Outline

The virial theorem

Simple models

Polytropes

Summary

Gravitational potential energy I

- The gravitational potential at distance r from the centre is $-G\,M(r)/r = -G\,m/r$
- Hence the total gravitational potential energy of the star, Ω:

$$\Omega = -\int_0^{M_*} \frac{G\,M(r)}{r}\,dM(r) = -\int_0^{M_*} \frac{G\,m}{r}\,dm$$

- Apply the mass equation:

$$\Omega = -4\pi \int_0^{R_*} r\,G\,M(r)\,\rho(r)\,dr = -4\pi \int_0^{R_*} r\,G\,m\,\rho\,dr$$

- Apply the equation of hydrostatic equilibrium:

$$\Omega = 4\pi \int_0^{R_*} \frac{dP}{dr} r^3\,dr$$

Gravitational potential energy II

- Integrate the RHS by parts:

$$\Omega = \Big[4\pi r^3\, P(r)\Big]_0^{R_*} - 3\int_0^{R_*} 4\pi r^2\, P(r)\, dr$$

- At the core, $r = 0$ and at the surface $P \to 0$, making the first term 0
- We can apply the mass equation to the second term:

$$\Omega = -3\int_0^{R_*} 4\pi r^2\, P(r)\, dr = -3\int_0^{M_*} \frac{P(r)}{\rho(r)}\, dM(r) = -3\int_0^{M_*} \frac{P}{\rho}\, dm$$

Virial theorem

- The virial theorem relates the total kinetic energy with the total potential energy of a star (Ω).
- We assume that the kinetic energy is the same as the total internal thermal energy (U).
- Split up between the non-relativistic and relativistic case.
- Remember that in an ideal gas, the internal thermal energy density is:

$$u_{non-rel} = \frac{3}{2}P$$

$$u_{rel} = 3P$$

Virial theorem - non-relativistic

- Assume an ideal gas of non-relativistic particles

$$\Omega = -3 \int_0^{R_*} 4\pi r^2 \, P(r) \, dr = -2 \int_0^{R_*} 4\pi \, r^2 \, u \, dr$$

- This is essentially an integral over the full volume of a star

$$\Omega = -2 \int_0^{R_*} u \, dV = -2 \, U$$

- Total energy of the star:

$$E = U + \Omega = -U = \frac{\Omega}{2}$$

- Note that $E < 0$. Star is stable!

Luminosity of a contracting star

- Globally, energy conservation is required. So, the energy "lost" by the star is radiated away $\rightarrow$ Luminosity - energy lost over time.
- This means that if a star contracts, i.e. gravitational energy is released (Ω decreases), half of the gravitational energy is transformed in thermal energy and the star heats up (U increases), whereas the other half must be radiated away.

$$L_G = -\frac{1}{2}\frac{d\Omega}{dt}$$

- So a star contracts and heats up as it radiates energy
- Selfgravitating systems have negative specific heat - as they lose energy they become warmer!
- Remember lecture 1 - this process happens on the Kelvin-Helmholtz timescale

Virial theorem - relativistic

- Assume an ideal gas of extremely relativistic particles

$$\Omega = -3 \int_0^{R_*} 4\pi r^2 \, P(r) \, dr = - \int_0^{R_*} 4\pi \, r^2 \, u \, dr$$

- This is essentially an integral over the full volume of a star

$$\Omega = - \int_0^{R_*} u \, dV = - U$$

- Total energy of the star:

$$E = U + \Omega = 0$$

- Extremely relativistic system is marginally stable - slightest change may cause instability.

Outline

The virial theorem

Simple models

Polytropes

Summary

Solving the equation of hydrostatic equilibrium

- Mass equation:

$$\frac{dM(r)}{dr} = \frac{dm}{dr} = 4\pi r^2 \rho(r)$$

- Equation of hydrostatic equilibrium:

$$\frac{dP(r)}{dr} = -\frac{GM(r)\rho(r)}{r^2}$$

- Not easily solvable. Notable exceptions:
  - Density is a known function of radius
  - Density is a known function of pressure
    - Isothermal atmospheres (see JCD 4.5.2)
    - Polytropes

Slido

Join SLIDO - #1114155

Exercises for known $\rho(r)$

- JCD 4.5.1 assumes a linear model: $\rho \propto r$. Verify equations 4.28, 4.29, 4.30.
- In a star with mass M_* and radius R_*, density decreases from the interior to the surface. This can be approximated with

$$\rho(r) = \rho_c \left[1 - \left(\frac{r}{R_*}\right)^2\right]$$

  where ρ_c is the central density.
  - Show that the average density of the star is equal to $0.4\rho_c$.

Outline

The virial theorem

Simple models

Polytropes

Summary

Polytropic models

- Polytropic models are models where the pressure and density are related as
$$P = K\rho^{\gamma}$$
with K and γ constant
- Special case: if $\gamma = 1$, we have an isothermal model.
- Other examples where polytropic models work include a gas where degenerate electrons dominate or when density and pressure are related adiabatically (see lecture 4).
- Convenient extra definitions / variables:
  - Polytropic index n: $n = \frac{1}{\gamma - 1}$
  - Dimensionless parameter θ such that $\rho = \rho_c \theta^n$
  - New "radius" measure ξ such that $r = \alpha\xi$ and $\alpha^2 = \frac{(n+1)K\,\rho_c^{1/n-1}}{4\pi\,G}$

Lane-Emden equation

- Using the polytropic model and the definitions above, it can be shown that
$$\frac{1}{\xi^2}\frac{d}{d\xi}\left(\xi^2\frac{d\theta}{d\xi}\right) = -\theta^n$$
- A polytropic star can be modeled if we can solve this Lane-Emden equation and find solutions $\theta(\xi)$ with boundary conditions $\theta(0) = 1$ (center) and $\theta(\xi_1) = 0$ at the surface.
- Analytical solutions only possible for $n = 0, 1, 5$ (see JCD 4.6 and exercise JCD 4.11). In all other case numerical solutions need to be found.

Summary

- Gravitational potential energy:

$$\Omega = -3 \int_0^{R_*} 4\pi r^2 \, P(r) \, dr = -3 \int_0^{M_*} \frac{P}{\rho} \, dm$$

- Virial theorem relates the total internal thermal energy (U) with the total potential energy of a star (Ω).
- Non-relativistic: $\Omega = -2\,U$
- Relativistic: $\Omega = -\,U$ (total energy is zero)
- Simple stellar models include
  - Density known function of radius
  - Isothermal atmospheres
  - Polytropic models. This leads to the Lane-Emden equation

NEXT LECTURE: Energy transport, radiative transport, mean free path